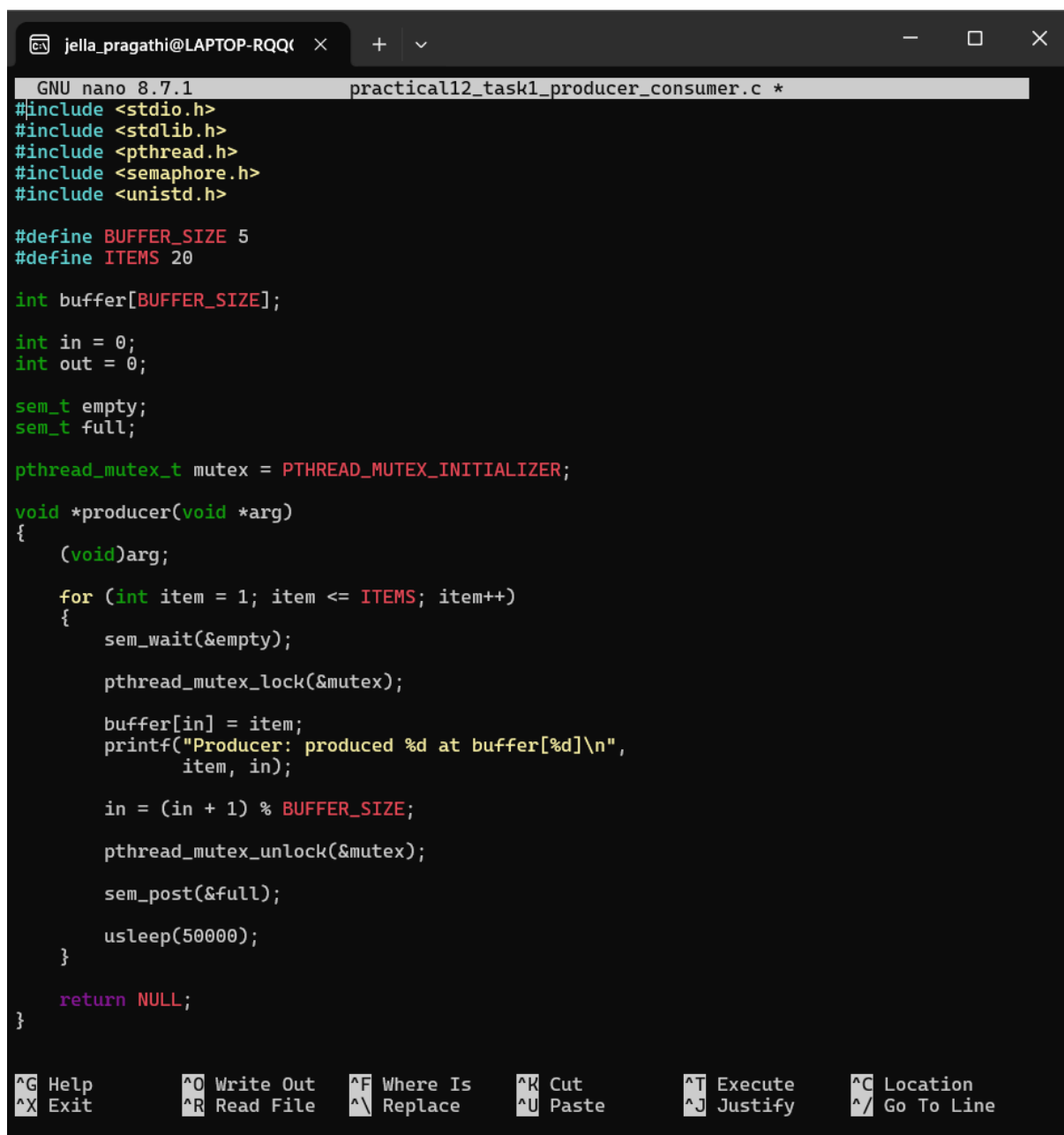


Practical Week 12

Session 12

Task 1

Implement the Producer-Consumer problem using counting semaphores and POSIX threads. Evaluate synchronization correctness and system throughput under different buffer sizes.



```
GNU nano 8.7.1 practical12_task1_producer_consumer.c *
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define BUFFER_SIZE 5
#define ITEMS 20

int buffer[BUFFER_SIZE];

int in = 0;
int out = 0;

sem_t empty;
sem_t full;

pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;

void *producer(void *arg)
{
    (void)arg;

    for (int item = 1; item <= ITEMS; item++)
    {
        sem_wait(&empty);

        pthread_mutex_lock(&mutex);

        buffer[in] = item;
        printf("Producer: produced %d at buffer[%d]\n",
            item, in);

        in = (in + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex);

        sem_post(&full);

        usleep(50000);
    }

    return NULL;
}
```

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```
jella_pragathi@LAPTOP-RQQC: ~$ nano practical12_task1_producer_consumer.c
GNU nano 8.7.1 practical12_task1_producer_consumer.c *
void *consumer(void *arg)
{
    (void)arg;

    for (int i = 0; i < ITEMS; i++)
    {
        sem_wait(&full);

        pthread_mutex_lock(&mutex);

        int item = buffer[out];

        printf("Consumer: consumed %d from buffer[%d]\n",
            item, out);

        out = (out + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex);

        sem_post(&empty);

        usleep(80000);
    }

    return NULL;
}

int main(void)
{
    pthread_t producer_thread;
    pthread_t consumer_thread;

    printf("=====\n");
    printf("Practical Session 12 - Task 1\n");
    printf("Producer-Consumer using Semaphores\n");
    printf("=====\n");

    printf("\nBuffer size: %d\n", BUFFER_SIZE);
    printf("Items: %d\n\n", ITEMS);

    if (sem_init(&empty, 0, BUFFER_SIZE) != 0)
    {
        perror("sem_init empty");
        return 1;
    }

    pthread_create(&producer_thread, NULL, producer, &mutex);
    pthread_create(&consumer_thread, NULL, consumer, &mutex);

    pthread_join(producer_thread, NULL);
    pthread_join(consumer_thread, NULL);

    printf("=====\n");
    printf("Practical Session 12 - Task 1\n");
    printf("Producer-Consumer using Semaphores\n");
    printf("=====\n");

    return 0;
}
```

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```

if (sem_init(&full, 0, 0) != 0)
{
    perror("sem_init full");
    sem_destroy(&empty);
    return 1;
}

if (pthread_create(&producer_thread, NULL,
                  producer, NULL) != 0)
{
    perror("pthread_create producer");
    return 1;
}

if (pthread_create(&consumer_thread, NULL,
                  consumer, NULL) != 0)
{
    perror("pthread_create consumer");
    return 1;
}

pthread_join(producer_thread, NULL);
pthread_join(consumer_thread, NULL);

sem_destroy(&empty);
sem_destroy(&full);
pthread_mutex_destroy(&mutex);

printf("\nProducer-consumer execution completed successfully.\n");

return 0;
}

```

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```
jella_pragathi@LAPTOP-RQQGT555:~$ nano practical12_task1_producer_consumer.c
jella_pragathi@LAPTOP-RQQGT555:~$ gcc -Wall -Wextra -pthread practical12_task1_producer_consumer.c -o practical12_task1_producer_consumer
jella_pragathi@LAPTOP-RQQGT555:~$ ./practical12_task1_producer_consumer
=====
Practical Session 12 - Task 1
Producer-Consumer using Semaphores
=====

Buffer size: 5
Items: 20

Producer: produced 1 at buffer[0]
Consumer: consumed 1 from buffer[0]
Producer: produced 2 at buffer[1]
Consumer: consumed 2 from buffer[1]
Producer: produced 3 at buffer[2]
Producer: produced 4 at buffer[3]
Consumer: consumed 3 from buffer[2]
Producer: produced 5 at buffer[4]
Consumer: consumed 4 from buffer[3]
Producer: produced 6 at buffer[0]
Producer: produced 7 at buffer[1]
Consumer: consumed 5 from buffer[4]
Producer: produced 8 at buffer[2]
Consumer: consumed 6 from buffer[0]
Producer: produced 9 at buffer[3]
Producer: produced 10 at buffer[4]
Consumer: consumed 7 from buffer[1]
Producer: produced 11 at buffer[0]
Producer: produced 12 at buffer[1]
Consumer: consumed 8 from buffer[2]
Producer: produced 13 at buffer[2]
Consumer: consumed 9 from buffer[3]
Producer: produced 14 at buffer[3]
Consumer: consumed 10 from buffer[4]
Producer: produced 15 at buffer[4]
Consumer: consumed 11 from buffer[0]
Producer: produced 16 at buffer[0]
Consumer: consumed 12 from buffer[1]
Producer: produced 17 at buffer[1]
Consumer: consumed 13 from buffer[2]
Producer: produced 18 at buffer[2]
Consumer: consumed 14 from buffer[3]
Producer: produced 19 at buffer[3]
Consumer: consumed 15 from buffer[4]
Producer: produced 20 at buffer[4]
Consumer: consumed 16 from buffer[0]
Consumer: consumed 17 from buffer[1]
```

```
jella_pragathi@LAPTOP-RQQG5555:~$ nano practical12_task1_producer_consumer.c
jella_pragathi@LAPTOP-RQQG5555:~$ gcc -Wall -Wextra -pthread practical12_task1_producer_consumer.c -o practical12_task1_producer_consumer
jella_pragathi@LAPTOP-RQQG5555:~$ ./practical12_task1_producer_consumer
=====
Practical Session 12 - Task 1
Producer-Consumer using Semaphores
=====

Buffer size: 10
Items: 20

Producer: produced 1 at buffer[0]
Consumer: consumed 1 from buffer[0]
Producer: produced 2 at buffer[1]
Consumer: consumed 2 from buffer[1]
Producer: produced 3 at buffer[2]
Producer: produced 4 at buffer[3]
Consumer: consumed 3 from buffer[2]
Producer: produced 5 at buffer[4]
Consumer: consumed 4 from buffer[3]
Producer: produced 6 at buffer[5]
Producer: produced 7 at buffer[6]
Consumer: consumed 5 from buffer[4]
Producer: produced 8 at buffer[7]
Consumer: consumed 6 from buffer[5]
Producer: produced 9 at buffer[8]
Producer: produced 10 at buffer[9]
Consumer: consumed 7 from buffer[6]
Producer: produced 11 at buffer[0]
Producer: produced 12 at buffer[1]
Consumer: consumed 8 from buffer[7]
Producer: produced 13 at buffer[2]
Consumer: consumed 9 from buffer[8]
Producer: produced 14 at buffer[3]
Producer: produced 15 at buffer[4]
Consumer: consumed 10 from buffer[9]
Producer: produced 16 at buffer[5]
Consumer: consumed 11 from buffer[0]
Producer: produced 17 at buffer[6]
Producer: produced 18 at buffer[7]
Consumer: consumed 12 from buffer[1]
Producer: produced 19 at buffer[8]
Producer: produced 20 at buffer[9]
Consumer: consumed 13 from buffer[2]
```

```
Producer: produced 13 at buffer[2]
Consumer: consumed 9 from buffer[8]
Producer: produced 14 at buffer[3]
Producer: produced 15 at buffer[4]
Consumer: consumed 10 from buffer[9]
Producer: produced 16 at buffer[5]
Consumer: consumed 11 from buffer[0]
Producer: produced 17 at buffer[6]
Producer: produced 18 at buffer[7]
Consumer: consumed 12 from buffer[1]
Producer: produced 19 at buffer[8]
Producer: produced 20 at buffer[9]
Consumer: consumed 13 from buffer[2]
Consumer: consumed 14 from buffer[3]
Consumer: consumed 15 from buffer[4]
Consumer: consumed 16 from buffer[5]
Consumer: consumed 17 from buffer[6]
Consumer: consumed 18 from buffer[7]
Consumer: consumed 19 from buffer[8]
Consumer: consumed 20 from buffer[9]

Producer-consumer execution completed successfully.
jella_pragathi@LAPTOP-RQQT555:~$
```

Task 2

Design a deadlock scenario involving multiple threads competing for shared resources. Demonstrate deadlock occurrence, identify the four necessary conditions for deadlock, and implement a deadlock prevention strategy using resource ordering.

```
jella_pragathi@LAPTOP-RQQ... x + v
GNU nano 8.7.1 practical12_task2_deadlock.c *
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

pthread_mutex_t resource1 = PTHREAD_MUTEX_INITIALIZER;
pthread_mutex_t resource2 = PTHREAD_MUTEX_INITIALIZER;

void *thread1_function(void *arg)
{
    (void)arg;

    printf("Thread 1: locking Resource 1...\n");
    pthread_mutex_lock(&resource1);

    printf("Thread 1: Resource 1 locked.\n");
    sleep(1);

    printf("Thread 1: waiting for Resource 2...\n");
    pthread_mutex_lock(&resource2);

    printf("Thread 1: Resource 2 locked.\n");

    pthread_mutex_unlock(&resource2);
    pthread_mutex_unlock(&resource1);

    return NULL;
}

void *thread2_function(void *arg)
{
    (void)arg;

    printf("Thread 2: locking Resource 2...\n");
    pthread_mutex_lock(&resource2);

    printf("Thread 2: Resource 2 locked.\n");
    sleep(1);

    printf("Thread 2: waiting for Resource 1...\n");
    pthread_mutex_lock(&resource1);

    printf("Thread 2: Resource 1 locked.\n");

    pthread_mutex_unlock(&resource1);
}

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```

```

printf("Thread 2: Resource 1 locked.\n");

pthread_mutex_unlock(&resource1);
pthread_mutex_unlock(&resource2);

return NULL;
}

int main(void)
{
    pthread_t thread1;
    pthread_t thread2;

    printf("=====\n");
    printf("Practical Session 12 - Task 2\n");
    printf("Deadlock Demonstration\n");
    printf("=====\n\n");

    pthread_create(&thread1, NULL, thread1_function, NULL);
    pthread_create(&thread2, NULL, thread2_function, NULL);

    pthread_join(thread1, NULL);
    pthread_join(thread2, NULL);

    pthread_mutex_destroy(&resource1);
    pthread_mutex_destroy(&resource2);

    printf("\nDeadlock demonstration completed.\n");

    return 0;
}

```

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jella_pragathi@LAPTOP-RQQ(x + v - □ ×

GNU nano 8.7.1 practical12_task2_ordered.c *

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

pthread_mutex_t resource1 = PTHREAD_MUTEX_INITIALIZER;
pthread_mutex_t resource2 = PTHREAD_MUTEX_INITIALIZER;

void *thread1_function(void *arg)
{
    (void)arg;

    printf("Thread 1: locking Resource 1...\n");
    pthread_mutex_lock(&resource1);

    printf("Thread 1: Resource 1 locked.\n");

    sleep(1);

    printf("Thread 1: locking Resource 2...\n");
    pthread_mutex_lock(&resource2);

    printf("Thread 1: Resource 2 locked.\n");
    printf("Thread 1: using both resources.\n");

    pthread_mutex_unlock(&resource2);
    pthread_mutex_unlock(&resource1);

    printf("Thread 1: resources released.\n");

    return NULL;
}

void *thread2_function(void *arg)
{
    (void)arg;

    printf("Thread 2: locking Resource 1...\n");
    pthread_mutex_lock(&resource1);

    printf("Thread 2: Resource 1 locked.\n");

    sleep(1);

    printf("Thread 2: locking Resource 2...\n");
```

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```

printf("Thread 2: locking Resource 2...\n");
pthread_mutex_lock(&resource2);

printf("Thread 2: Resource 2 locked.\n");
printf("Thread 2: using both resources.\n");

pthread_mutex_unlock(&resource2);
pthread_mutex_unlock(&resource1);

printf("Thread 2: resources released.\n");

return NULL;
}

int main(void)
{
    pthread_t thread1;
    pthread_t thread2;

    printf("=====\n");
    printf("Practical Session 12 - Task 2\n");
    printf("Deadlock Prevention - Resource Ordering\n");
    printf("=====\n\n");

    pthread_create(&thread1, NULL, thread1_function, NULL);
    pthread_create(&thread2, NULL, thread2_function, NULL);

    pthread_join(thread1, NULL);
    pthread_join(thread2, NULL);

    pthread_mutex_destroy(&resource1);
    pthread_mutex_destroy(&resource2);

    printf("\nDeadlock prevented successfully.\n");

    return 0;
}

```

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```

Producer-consumer execution completed successfully.
jella_pragathi@LAPTOP-RQQT555:~$ nano practical12_task2_deadlock.c
jella_pragathi@LAPTOP-RQQT555:~$ gcc -Wall -Wextra -pthread practical12_task2_deadlock.c -o practical12_task2_deadlock
jella_pragathi@LAPTOP-RQQT555:~$ ./practical12_task2_deadlock
=====
Practical Session 12 - Task 2
Deadlock Demonstration
=====

Thread 1: locking Resource 1...
Thread 1: Resource 1 locked.
Thread 2: locking Resource 2...
Thread 2: Resource 2 locked.
Thread 1: waiting for Resource 2...
Thread 2: waiting for Resource 1...
^C
jella_pragathi@LAPTOP-RQQT555:~$ ^C
jella_pragathi@LAPTOP-RQQT555:~$ nano practical12_task2_ordered.c
jella_pragathi@LAPTOP-RQQT555:~$ gcc -Wall -Wextra -pthread practical12_task2_ordered.c -o practical12_task2_ordered
jella_pragathi@LAPTOP-RQQT555:~$ ./practical12_task2_ordered
=====
Practical Session 12 - Task 2
Deadlock Prevention - Resource Ordering
=====

Thread 1: locking Resource 1...
Thread 1: Resource 1 locked.
Thread 2: locking Resource 1...
Thread 1: locking Resource 2...
Thread 1: Resource 2 locked.
Thread 1: using both resources.
Thread 1: resources released.
Thread 2: Resource 1 locked.
Thread 2: locking Resource 2...
Thread 2: Resource 2 locked.
Thread 2: using both resources.
Thread 2: resources released.

Deadlock prevented successfully.
jella_pragathi@LAPTOP-RQQT555:~$

```